

Homework 1

You can submit the homework as *.py or *.ipynb file. Answers to theoretical questions can be written in a separate WORD document.

- 1.) Setup a python environment on your computer (Pycharm / google colab).
- 2.) Install the following libraries: pytorch, NLTK, spaCy
- 3.) Write a program that loads the file spam.csv (source: KAGGLE)
- 4.) Compute simple statistics on the data: number of SMS message, number of spams, total word count, averaged number of words per message, 5 most frequent words in the file, number of rare words (appear only once in the file).
- 5.) Perform tokenization twice: by using NLTK library and by Spacy library. Write what is the difference between the results.
- 6.) Perform lemmatization twice: by NLTK and by Spacy libraries. Write what is the difference between the results.
- 7.) Perform stemming twice (NLTK and Spacy). Write what is the difference between the results.
- 8.) Find a message that if the analysis will be repeated on the spam.csv without it, then there will be less stemmed tokens but number of lemmatized tokens will not be changed. If you do not find such a message – explain why.
- 9.) Find a message that if the analysis will be repeated on the spam.csv without it, then there will be less lemmatized tokens but the number of stemmed tokens will not be changed. If you do not find such a message – explain why.
- 10.) Compare the results of NLTK vs. Spacy.
- 11.) Compare the results of NLTK before and after tokenization, lemmatization and stemming
- 12.) Compare the results of Spacy before and after tokenization, lemmatization and stemming
- 13.) Install the library BeautifulSoup
- 14.) Choose any url page. You can choose a page of yourself in a social media (e.g., facebook, Instagram, Linkdin, Twitter), or anything else
- 15.) Perform data scrapping of this url. You can learn how to do it from the following links:
<https://www.edureka.co/blog/web-scraping-with-python/>
<https://zindi.africa/learn/a-beginners-guide-to-scraping-data-from-social-media>
or, ask CHATGPT to program it for you.
- 16.) Perform steps 5-11 on the text achieved in (14).
- 17.) Download one of your whatsapp chats into *.txt file (at least 50 messages in Hebrew).
- 18.) Repeat steps 5-11 on the text file achieved in (16)